Aspect-Level Paper 1 Commitments: Formalizing How Each Paper 1 Governance Commitment Applies at Aspect Scope

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 12, 2026

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026).

Abstract

The Coordination Knowledge Substrate (CKS) pattern commits, through Paper 2's recursive-levels principle, to Paper 1 governance holding at every architectural level — cell, aspect, and Self. B1.20 establishes this commitment at source; B2.98 frames the recursive application across levels; B2.99 formalizes cell-level application. This note, B2.100, formalizes *aspect-level* application: what each Paper 1 governance commitment means concretely when applied at aspect scope. The aspect is a coordination entity — a purpose-defined arrangement of cells — and its commitments therefore govern coordination behavior rather than task behavior. A1.01 at aspect scope means humans govern how aspects coordinate cells through aspect DNA; A1.03 means coordination conflicts within the aspect are registered as first-class substrate state; A1.04 means the aspect's coordination substrate mediates between cell operations and aspect-level outputs, a different concrete referent than instinct/reasoning mediation at cell scope; and so on through A2.46. The note specifies the full set, articulates what makes aspect-scope application architecturally distinctive from cell-scope application, draws the biological analog of tissue-level regulatory governance, states operational implications, identifies limits, and provides an operational test. B2.19 (aspect-level inheritance verification) is the verification counterpart; B2.100 is the specification counterpart.

1. Why aspect-level Paper 1 commitments requires standalone formalization

Paper 2 commits to Paper 1 governance at every level. That commitment is easy to state in a single sentence — "Paper 1 commitments apply recursively at cell, aspect, and Self scope" — and considerably harder to act on without knowing what each commitment means concretely at each scope. The meaning is not obvious: A1.04 at cell scope concerns the relationship between the LLM instinct layer and the CKS reasoning substrate; A1.04 at aspect scope concerns the relationship between the coordination substrate and the cells it coordinates. These are the same abstract commitment — AI as substrate mediator — but the concrete referents differ by scope. Without formalization, the recursive-levels principle is an architectural declaration rather than an operational specification.

B2.98 provides the integrating frame: Paper 1 commitments apply at every level via recursive scoping, with level-specific concrete meanings for each. B2.99 supplies the cell-level specification. B2.100 supplies the aspect-level specification. B2.101 will supply the Self-level specification. Together, these three notes translate the recursive commitment from architectural declaration into operational specification at each scope.

The aspect-level specification is the coordination-scope specification. An aspect coordinates cells toward a defined purpose; its commitments govern how that coordination is structured, governed, and made accountable. This is architecturally distinct from cell-level commitments, which govern task execution within a cell, and from Self-level commitments, which govern the integrated whole. The aspect occupies the middle coordination layer, and Paper 1 commitments at that layer have coordination-specific meanings that require explicit statement.

The defensive-publication function of this note is to establish prior art at the level of granularity where aspect-scope commitments could otherwise be claimed as novel invention. Prior art at the recursive-commitment level — established at B1.20 — does not by itself constitute prior art at each level's concrete application. The level-specific formalization is the prior art this note creates. As the one-hundredth Phase B2 note and the third of thirteen decomposing B1.20, B2.100 fills the coordination-scope specification gap in the prior-art chain.

2. The architectural specification: each Paper 1 commitment at aspect scope

Aspect scope, per B1.18 and B2.15–B2.16, covers the coordination domain defined by the aspect's purpose: the set of cells the aspect coordinates, the coordination rules governing how those cells contribute to aspect-level outputs, and the aspect's structural position within the Self. Aspect DNA, per B2.16, is the substrate carrying coordination rules — membership rules specifying which cells participate, invocation rules specifying how and when cells are called, output-integration rules specifying how cell contributions combine, conflict-handling rules specifying how coordination conflicts are addressed, and purpose-alignment rules specifying how coordination behavior serves the aspect's purpose. Aspect lineage, per B2.43, is the operational history of the aspect from birth through its current state.

With that scope established, each Paper 1 commitment applies as follows.

A1.01 — Human-governed at aspect scope. The aspect's coordination behavior is human-governed: humans hold the rights to inspect, modify, and override aspect DNA at any time. Aspect governance means humans govern how aspects coordinate cells — what cells are members, how they are invoked, how their outputs are integrated, how conflicts are handled, and how coordination aligns with the aspect's purpose. This is governance at coordination scope: humans govern each aspect's coordination behavior specifically, not only the deployment as a whole. The actor-neutrality from Paper 1 carries through: human-governed at aspect scope does not mean humans author every cell output; it means humans hold authority over the coordination rules that govern how cell outputs are organized and combined.

A1.03 — Conflict-as-first-class at aspect scope. Conflicts arising within aspect coordination — between cells producing inconsistent outputs, between coordination rules producing inconsistent outcomes, between purpose-alignment rules and invocation outcomes — are preserved as first-class substrate state rather than silently resolved. Aspect DNA includes explicit conflict-handling rules per B2.16; unresolved coordination conflicts are registered for governance rather than discarded or implicitly collapsed. The conflict-preservation commitment at aspect scope is coordination conflict preservation: the aspect maintains a record of coordination-layer disagreements that the cell level did not and the Self level need not subsume.

A1.04 — AI-as-substrate-mediator at aspect scope. At aspect scope, A1.04's mediation role is performed by the aspect's coordination substrate — the aspect DNA — rather than by the LLM directly. Aspect DNA mediates between cell operations and aspect-level outputs: coordination rules specify how cell contributions combine, in what order cells are invoked, and how output-integration proceeds. The aspect is the substrate-mediated coordinator. This is architecturally distinct from A1.04 at cell scope, where the CKS substrate mediates between the LLM instinct layer and coordinated human-governed reasoning. At aspect scope, mediation is coordination mediation, not instinct/reasoning mediation. The abstract commitment — a governed substrate mediates between lower-level operations and higher-level outputs — holds; the concrete mediation relationship is coordination-specific.

A1.05 — Tool-agnosticism at aspect scope. Aspect coordination behavior is specified in aspect substrate regardless of which specific cells are member cells or what implementations those cells use. Aspect coordination rules per B2.16 apply to member cells regardless of those cells' specific LLM models, substrate implementations, or infrastructure configurations. The aspect binds to coordination roles, not to specific cell implementations. Cells satisfying the aspect's membership and invocation rules participate in coordination regardless of their underlying tooling, and the aspect's coordination behavior is determined by its DNA, not by the particular tools cells happen to use.

A1.07 — Retraceability at aspect scope. Aspect operations are retraceable through aspect lineage per B2.43. The aspect's coordination history — which cells were members at each state, which coordination rules were applied, what coordination decisions were made, how aspect DNA evolved over time — is retraceable from aspect birth through current state. Aspect-level operational accountability is available at the aspect's own scope without requiring traversal of the full Self or cell-level histories. Aspect lineage is the retraceability substrate for coordination-level accountability.

A1.08 — Substrate-as-source-of-truth at aspect scope. Aspect substrate — aspect DNA per B2.16 and aspect membership per B2.08 — is the authoritative source of truth for aspect coordination behavior. What the aspect does in coordination is determined by what its substrate specifies, not by LLM inference about how coordination should work. If aspect coordination behavior is in question, the authoritative answer is in the aspect substrate. This is the same commitment as at cell scope — substrate, not inference, is authoritative — applied to the coordination substrate rather than to the task substrate.

A1.10 — Determinism at aspect scope. Given aspect coordination rules and the behaviors of member cells, aspect coordination outcomes are deterministic. The same aspect DNA applied over the same cell behaviors produces the same aspect-level coordination outcome. Aspect-level determinism is coordination determinism: the coordination function is determined by substrate content, not by runtime inference about how to coordinate. Variation in aspect coordination outcomes traces to variation in aspect DNA or in cell behavior, not to non-deterministic coordination logic operating outside the substrate.

A1.12 — Labor allocation at aspect scope. Cells perform coordination labor within the aspect: they execute cell operations that contribute to aspect-level outputs under the coordination rules aspect DNA specifies. LLMs may perform cell-level operations that flow into aspect coordination. Humans govern aspect DNA — authoring coordination rules, modifying membership, overriding coordination decisions — while cells execute the coordination work the rules direct. Authority over coordination structure is human; the labor of coordinated task execution is cell-executed. Labor is allocable; authority over the coordination rules is not.

A1.13 — Composition requirements at aspect scope. Aspects satisfy composition requirements within Selves per A5.14: aspects compose into a Self under governance requirements the Self specifies, and an aspect that fails to satisfy those requirements is not a valid Self component. Aspects also impose composition requirements on their member cells: cells must satisfy the aspect's membership rules, invocation-compatibility rules, and output-compatibility rules to serve as valid aspect members. The composition requirement relationship is bidirectional — aspects are required of by the Self above and require of cells below.

A2.01–A2.04 — Governance affordances at aspect scope. The four governance affordances apply at aspect granularity. A2.01 (inspect): humans inspect aspect DNA, aspect membership, and aspect coordination history. A2.02 (modify): humans modify aspect coordination rules, amending DNA content under the modification right at any time. A2.03 (override): humans override specific aspect coordination decisions, exercising the override right at coordination-decision granularity rather than only at deployment or cell level. A2.04 (rule authoring): humans author aspect coordination rules — the membership rules, invocation rules, output-integration rules, conflict-handling rules, and purpose-alignment rules that constitute aspect DNA. Rule authoring is the primary governance mechanism at aspect scope: the coordination rules humans author determine what the aspect does.

A2.40 — Provenance at aspect scope. Aspect coordination events record provenance: which coordination rules were applied, which cells contributed, which coordination decisions were made, and under whose authority modifications to aspect DNA were made. Coordination-level provenance is available independently of cell-level provenance, providing an aspect-granularity accountability record.

A2.46 — Authoritative content at aspect scope. Aspect DNA is Category 4 authoritative content: it specifies what coordination rules govern this aspect. Aspect DNA is not advisory coordination guidance; it is the authoritative specification of how this aspect coordinates its member cells. Coordination behavior is determined by this content, and the content's authority derives from its status as human-governed substrate rather than from any runtime inference that might otherwise characterize how coordination should proceed.

3. What makes aspect-level commitments architecturally distinctive

The aspect-level application is architecturally distinctive in one respect above all others: it is coordination governance, not task governance.

At cell scope, Paper 1 commitments govern task behavior — how a cell executes its informational task, how LLM inference is mediated by the CKS substrate, how cell-level conflicts are preserved and addressed. At aspect scope, the same commitments govern coordination behavior — how cells are organized and invoked toward an aspect's purpose, how coordination conflicts are preserved, how coordination substrate mediates between cells and aspect outputs. The commitment structure is inherited intact; the referent shifts from task execution to coordination.

This shift has concrete architectural consequences. A1.04 at cell scope names the LLM-as-mediated-by-substrate as the mediation relationship; A1.04 at aspect scope names aspect-DNA-as-mediating-coordination as the mediation relationship. A1.01 at cell scope names human authority over cell substrate and orchestration rules; A1.01 at aspect scope names human authority over the coordination rules governing how cells are organized. The abstract commitment — human authority, substrate mediation, conflict registration, retraceability — applies unchanged; the concrete entity it applies to is the aspect's coordination substrate, not a cell's task substrate.

This is precisely the architectural advantage of the recursive-levels principle: the same abstract commitment set applies at every scope with level-specific concrete meanings, rather than requiring a distinct governance framework at each level. The aspect does not require a new kind of governance; it requires the same governance applied at coordination scope. The prior-art significance is that the concrete coordination-scope application of each commitment is specified here, not merely asserted as following from the recursive principle.

4. The biological analog

In biological systems, tissues do not replicate the molecular-scale regulatory mechanisms that govern individual cells; they operate at their own regulatory scale. Tissue-level regulation governs how cells contribute to tissue function — which cells are recruited, how their outputs are integrated, how the tissue responds to signals from the organism level above — without duplicating or replacing cell-level regulatory mechanisms. The same biological principles of regulatory signal, feedback, and homeostasis apply at tissue scope with tissue-appropriate concrete referents.

The CKS aspect is the governed architectural analog. Aspect-level commitments apply Paper 1 principles at coordination scope — with aspect-appropriate concrete referents — without duplicating or replacing cell-level commitments. A1.04 at tissue scope, if formalized in biological terms, would govern how the tissue's signaling substrate mediates between cell contributions and tissue-level outputs; A1.04 at aspect scope governs how the aspect's coordination substrate mediates between cell contributions and aspect-level outputs. The structural parallel is exact. The governed architectural instantiation exceeds the biological analog by being explicitly specified as substrate content under human authority rather than emerging through evolutionary regulatory layering — this is the B1.20 excess-over-biology point applied to the governance dimension specifically.

The biological analog is load-bearing in the same bounded sense Paper 2's biology-as-bounded-terminology framing establishes: it provides conceptual scaffolding for the multi-level governance architecture without committing to biological connotations — evolutionary pressure, fitness landscapes, stochastic mutation — that do not apply at the architectural-pattern level.

5. Inherited Paper 1 commitments

All Paper 1 commitments apply at aspect scope. Section 2 specifies what aspect-scope application means for each commitment enumerated there. The specification is not exhaustive of every nuance in the Series A foundational commitment notes; it identifies the aspect-scope concrete meaning for each commitment in a form that operationalizes the recursive principle.

B2.19 (aspect-level inheritance verification) is the verification counterpart to this specification. Where B2.100 states what each commitment means at aspect scope, B2.19 provides the verification framework for confirming that a given aspect satisfies those commitments. The specification-verification pair is the complete treatment of aspect-level Paper 1 inheritance: B2.100 answers what commitments mean here; B2.19 answers how to confirm they are satisfied.

Series A foundational commitment notes provide the detailed formalization of each individual commitment at base register. B2.100 does not repeat that formalization; it applies it at aspect scope. The aspect-scope meaning is derived from, and inherits from, the Series A formalization. Where a Series A note establishes, for example, that A1.04 names a substrate-mediated relationship between lower-level operations and higher-level outputs, B2.100 specifies that at aspect scope the lower-level operations are cell operations and the higher-level output is the aspect-level coordination outcome, with aspect DNA as the mediating substrate. The inheritance is derivational, not assertional.

6. Operational implications

Several operational implications follow directly from the aspect-level specification.

Each aspect in a deployment satisfies Paper 1 commitments at its own scope independently of every other aspect. This is not a deployment-level aggregate satisfaction but a per-aspect satisfaction: each aspect's coordination substrate is human-governed, each aspect registers coordination conflicts as first-class, each aspect's substrate is authoritative for its coordination behavior. Compliance demonstrations are therefore possible at aspect granularity rather than only at cell or deployment level, enabling governance reviews and accountability assessments to be scoped to individual coordination arrangements rather than requiring full-deployment analysis.

Aspect-level governance reviews focus on coordination behavior as their primary subject matter. A governance review of an aspect examines aspect DNA — whether coordination rules are human-authored, whether conflict-handling rules are explicit and complete, whether membership and invocation rules are correctly specified, whether the aspect's provenance records are intact — not cell-level task behavior. This is a distinct review scope that becomes available precisely because aspect-scope commitments are specified at coordination granularity.

Aspect DNA authoring per A2.04 is the primary governance mechanism for aspect-level commitment application. The coordination rules governing an aspect are authored by humans at design time and updated through the modification right at any subsequent time without architectural gating. Governance is exercised primarily at rule-authoring scope, with override as the intervention mechanism for specific coordination decisions that deviate from governed expectation. The cost structure of aspect-level governance parallels the cost structure Paper 1 establishes for cell-level governance: rule authoring amortizes across every coordination execution that follows; override cost is proportional to intervention frequency, not to aspect size or cell count.

Aspect lineage per B2.43 provides the operational history for aspect-level retraceability per A1.07. An aspect's coordination accountability chain — from birth through every coordination decision, every modification to aspect DNA, every change in cell membership — is available through lineage traversal at aspect scope. This is distinct from cell lineage and from Self lineage; aspect-level accountability is available independently at its own scope and does not require traversal of the full Self history to establish coordination-level accountability.

7. Limits

Aspect-level commitments do not duplicate cell-level commitments. The same Paper 1 principles appear at both scopes, but they govern different objects: cell-level commitments govern task behavior; aspect-level commitments govern coordination behavior. The scope distinction is real and architecturally load-bearing. An aspect satisfying A1.04 at coordination scope and a cell satisfying A1.04 at task scope are both satisfying the same abstract commitment, but the concrete satisfaction conditions differ by scope. Confusing the scopes — treating aspect-level compliance as a substitute for cell-level compliance or vice versa — produces misapplication of both.

Aspect-level governance does not replace deployment-level or cell-level governance. All three governance scopes are active simultaneously in a CKS deployment: each cell is governed at cell scope, each aspect is governed at aspect scope, and the Self is governed at Self scope. These are parallel governance operations rather than hierarchical substitutes. An aspect satisfying its aspect-scope commitments does not substitute for its member cells satisfying cell-scope commitments, nor does it substitute for the Self satisfying Self-scope commitments. Each scope's governance is necessary; none is sufficient for the others.

B2.19 is the verification counterpart to B2.100; B2.100 is not a verification framework. This note specifies what commitments mean at aspect scope; B2.19 provides the mechanisms for verifying satisfaction. A deployment seeking to demonstrate aspect-level Paper 1 compliance looks to B2.19 for the verification apparatus and to B2.100 for the specification of what is being verified. The notes are complementary, not redundant.

This note does not propose specific aspect governance products, coordination-level compliance frameworks, or mid-tier audit tools. The prior art established is at the architectural-specification level: what the commitments are and what they mean at aspect scope. Downstream work may instantiate specific products, tools, or compliance frameworks that operate over this specification; that downstream work is outside the scope of the prior-art chain this note contributes to.

8. Operational test

An aspect instantiates Paper 1 commitments at aspect scope if and only if all of the following hold at all times during the aspect's existence:

1. Humans hold the rights to inspect, modify, and override aspect DNA at any time, without architectural gating by any LLM operation, runtime middleware, or vendor policy.
2. Aspect DNA contains explicit, human-authored coordination rules covering at minimum: membership rules, invocation rules, output-integration rules, conflict-handling rules, and purpose-alignment rules.
3. Coordination conflicts arising within the aspect — between cell outputs, between coordination rules, between purpose-alignment and invocation outcomes — are preserved as first-class substrate state and

registered for governance rather than silently resolved.

4. The aspect's coordination substrate — aspect DNA — is the authoritative source of coordination behavior for this aspect; LLM inference does not substitute for aspect substrate specification as the operative coordination specification.

5. Aspect coordination events record provenance through aspect lineage, maintaining an operational history retraceable from birth through current state.

6. Cells perform coordination labor within the aspect; humans govern the coordination rules that determine how that labor is organized, invoked, and integrated.

7. The aspect satisfies composition requirements within the Self that contains it, and member cells satisfy the aspect's composition requirements for participation.

An aspect that fails any of (1)–(7) may coordinate cells usefully but does not satisfy Paper 1 commitments at aspect scope in the sense this note formalizes.

9. Why naming as standalone matters

This is the third note in the thirteen-note B1.20 decomposition. B2.98 established the recursive commitment integrating frame: Paper 1 applies at every level via recursive scoping, with each level's concrete meanings derivable from the abstract commitments by substituting level-appropriate referents. B2.99 supplied the cell-level specification, establishing what each commitment means when the scope object is a cell executing an informational task. B2.100 supplies the aspect-level specification, establishing what each commitment means when the scope object is an aspect coordinating cells toward a purpose. The next notes will supply the Self-level specification (B2.101), per-commitment recursive formalizations that develop individual commitments across all three levels (B2.102–B2.108), operational tests for the full recursive commitment stack (B2.109), and the verification framework that closes Phase B2 (B2.110).

Without the aspect-level specification as a standalone note, the recursive-levels commitment is formally incomplete: one can assert that Paper 1 applies at aspect scope but cannot specify what that application means for any individual commitment. That incompleteness is precisely where subsequent patent claims could locate novelty — claiming aspect-level application of A1.04, or A1.03, or A2.04 as novel invention not covered by existing prior art. This note closes that gap for all Paper 1 commitments simultaneously, at the level of concrete specification rather than abstract assertion. The prior-art chain established here covers the full coordination-scope application of every Paper 1 commitment in a single prior-art record.

The note is the one-hundredth in Phase B2 and the three-hundred-seventeenth in the full derivation series. Its position in the B1.20 decomposition places it as the coordination-scope specification note — the note that makes aspect governance operationally meaningful by specifying what governed coordination behavior consists of at each commitment's scope.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model.* April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Aspect-Level Paper 1 Commitments: Formalizing How Each Paper 1 Governance Commitment Applies at Aspect Scope.* May 12, 2026. ORCID: 0009-0004-8065-3235.